AI-assisted inventory system for FAN Restaurant

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Abstract

This thesis investigates whether an AI-assisted inventory monitoring system can reduce manual work, improve record accuracy, and support more transparent stock management in a small restaurant environment. The project is based on a case study of FAN, a London-based restaurant whose outgoing-stock process relies on staff manually searching for inventory items, entering quantities, and uploading supporting photos. The research used semi-structured interviews, field observations, and iterative prototyping to understand the restaurant's workflow and translate practical problems into system requirements.

Several prototype iterations were developed and evaluated against the restaurant context. An initial vision-language model approach tested whether a single tray image could be matched against the full inventory list, but the response time was often longer than 35 seconds and the recognition results were unstable for visually similar products. A local YOLO-based proof of concept improved speed and reduced dependence on internet access, but it required labelled training images for each stock keeping unit, making it difficult to scale across the restaurant's wider inventory. Later iterations therefore used narrower kitchen and bar contexts, recent stock movement data, and a human confirmation step to reduce the recognition task. The final design became a chat-based inventory agent that followed the kitchen team's existing habit of sharing stock photos through WhatsApp-like messaging.

The findings suggest that AI can support restaurant inventory recording when it is designed around existing staff behaviour rather than added as a separate technical task. A general-purpose vision-language model provides a flexible approach to item recognition, but its performance depends on constrained candidate lists, human confirmation, image quality, and the completeness of the SKU database. The final evaluation showed that the system performed well across the tested image recognition scenarios, with single-image recognition generally completed within one second. However, further work is needed to evaluate the system through longer-term staff use and across different restaurant environments. Future development could also introduce CCTV-based verification as an additional double-check mechanism and improve operational practices through better SKU maintenance and staff guidance.

Acronyms

AI Artificial Intelligence

BTOS Business Trends and Outlook Survey

CCTV Closed-Circuit Television

CNN Convolutional Neural Network

CV Computer Vision

ICT Information and Communication Technology

JSON JavaScript Object Notation

MVP Minimum Viable Product

OECD Organization for Economic Co-operation and Development

RGB-D Red-Green-Blue-Depth

RFID Radio-Frequency Identification

RPC Retail Product Checkout

RSSI Received Signal Strength Indicator

SKU Stock Keeping Unit

UHF Ultra-High Frequency

VLM Vision-Language Model

YOLO You Only Look Once

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And finally, while I am writing this thesis, young people my age in Palestine are facing starvation and blockade, not knowing where their next meal will come from or whether their homes will be bombed. What is happening is inhumane. May there be peace in Palestine and around the world.

Declaration

This thesis has been created by myself and has not been submitted in any previous application for any degree. The work in this thesis has been undertaken by myself except where otherwise stated.

1 Introduction

Artificial intelligence has the potential to improve efficiency in the restaurant industry, an area where daily operations still rely heavily on manual recording, repetitive check-ups, and staff experience [1]. However, AI adoption has not developed evenly across sectors. According to the Organization for Economic Co-operation and Development (OECD), the use of AI in 2024 was much higher in the Information and Communication Technology (ICT) sector, reaching almost 44%, while accommodation and food services remained much lower at 7.8% [2]. Similarly, data from the U.S. Census Bureau's Business Trends and Outlook Survey (BTOS) show substantially higher AI adoption in the Information (39.7%) and Finance and Insurance (33.9%) sectors than in Retail Trade (approximately 14%) [3]. These figures suggest that physical service industries, including restaurants, are still relatively behind in AI-driven operational transformation.

This gap motivated this project to explore whether AI could provide practical value in a real restaurant environment. Instead of starting from a purely technical problem, a semi-structured interview was conducted with the owner of FAN [4], a London-based fine dining restaurant combining Japanese and Peruvian cuisine, to understand the operational difficulties involved in daily inventory management. The interview showed that inventory recording is still largely dependent on staff memory, manual input, and occasional checking, which can lead to missing records, incorrect quantities, and limited stock visibility. Based on this need, the research questions are:

RQ1: **How can an AI-assisted inventory monitoring system be designed to reduce manual work in a restaurant environment?**

RQ2: **How can an AI-assisted inventory monitoring system be designed to improve record accuracy in a restaurant environment?**

RQ3: **How can an AI-assisted inventory monitoring system be designed to support more transparent stock management in a restaurant environment?**

The project also has a broader motivation beyond solving one restaurant's immediate problem. If low-friction AI-assisted approaches can support inventory recording and monitoring in busy restaurant environments, they may provide a practical route for improving stock visibility without adding significant extra work for staff. There-

fore, this project aims to explore whether AI can support inventory recording in a real restaurant environment by reducing manual input and improving the visibility of stock movements.

2 Literature Review

To understand how an automated restaurant inventory monitoring system could be designed to reduce manual work, improve record accuracy, and support more transparent stock management, this section reviews existing research across three related computer science domains: inventory monitoring applications, visual recognition models, and supporting hardware and sensor systems. These three domains were selected because they map onto the main design decisions in this project: the inventory workflow defines the operational problem, visual recognition models determine how items can be identified, and hardware or sensor systems affect whether the solution can be deployed reliably in a restaurant storage environment.

2.1 Computer Vision Applications in Retail and Restaurant Inventory Systems

Muñoz et al. [5] built a supermarket shelf monitoring system for beverage stock detection. They used fixed Raspberry Pi cameras [6] to take shelf images, then applied image correction and cropping before sending the images into the model. Their system first used a small convolutional neural network (CNN) to check whether the image was usable, for example whether the shelf was blocked. If the image was valid, a YOLOv2 model [7] detected three beverage types and estimated stock levels. The result was then sent to a database and shown in a mobile app. This paper is relevant to this project because it shows that Raspberry Pi cameras can provide a practical and visually tidy solution for fixed-location inventory monitoring in storage areas. Compared with using a full computer setup in a warehouse, Raspberry Pi-based cameras are smaller, cheaper, and easier to place near shelves.

Gonzalez et al. [8] designed a tray-based food monitoring system in a dining hall. They placed a Red-Green-Blue-Depth (RGB-D) camera above the tray to capture food images and depth information. In contrast to the YOLOv2 model used by Muñoz et al., which mainly detected beverage categories through bounding

boxes, Gonzalez et al. used YOLOv8L-Seg, a segmentation version of YOLOv8 [9], to identify food areas at the pixel level. The depth data was then used to estimate food volume and weight. This is useful for this project because it shows that tray-based camera monitoring is possible in a restaurant-like setting. However, their system is more complex than this project's system because it estimates food weight, while this project's Minimum Viable Product (MVP) mainly focuses on recognising and verifying inventory items on a tray.

Tan et al. [10] proposed an improved YOLOv10 model for retail self-checkout, where the system detects multiple products placed together on a checkout counter. Their model was trained and evaluated using 30,000 checkout images from the Retail Product Checkout (RPC) dataset [11] and achieved strong detection performance. This paper is relevant to this project because this system also needs to recognise multiple inventory items placed together on a tray. The scale of their dataset shows that YOLO-based systems require large computing resources, data storage, and manual labour for annotation, especially when the system needs to distinguish between visually similar retail products. This is relevant to small restaurant settings because it raises the question of whether high-performing systems are practical when they depend on extensive labelled datasets and substantial computational resources. In contrast, this project uses a more responsible lightweight solution with lower computing requirements and no need for large-scale manual annotation. Rather than training a custom object detection model from scratch, the project explores the use of general-purpose vision-language models to support usable inventory recognition in a small restaurant context.

Overall, these studies show that computer vision can support inventory and product recognition tasks when the camera position and environment are controlled. However, most existing studies focus on supermarket shelves, checkout counters, or food portion estimation, and they often require large labelled datasets for model training and evaluation. In contrast, this project focuses on a smaller restaurant inventory setting, where the aim is to support stock recording without requiring large-scale data collection or manual annotation.

2.2 Vision-Based Product Recognition: YOLO Models and General-Purpose AI Models

Recent computer vision and vision-language model studies show two main directions for vision-based product recognition: task-specific object detection models such as YOLO, and more flexible vision-language models that can understand both images and text. These two approaches are relevant to this project because the system needs to identify inventory items from tray images, while also balancing speed, accuracy, cost, and ease of deployment.

Wang et al. [12] introduced YOLOv10, a real-time object detection model designed to improve both speed and accuracy. A key technical improvement for this version of YOLO is its attempt to remove the need for traditional non-maximum suppression, which normally adds extra post-processing time after detection. This makes YOLOv10 more suitable for real-time applications. For this project, YOLOv10 is relevant because inventory recognition may eventually need to run quickly on a local camera setup. However, YOLOv10 still mainly works as a closed-set detector, meaning it can only reliably detect categories that have been included in training.

Cheng et al. [13] proposed YOLO-World, which extends YOLO into open-vocabulary object detection. Unlike normal YOLO models, YOLO-World can use text prompts to detect objects that are not fixed in a predefined class list. It combines YOLO-style detection with vision-language pretraining, allowing it to detect objects in a zero-shot manner. This is useful for inventory systems because it may reduce the need to fully train every stock keeping unit (SKU) from scratch. However, for fine-grained product recognition, such as distinguishing between similar bottles or brands, open-vocabulary detection may still be less reliable than a custom-trained SKU model because product identity often depends on small packaging, label, flavour, and brand differences that are not always captured by broad text prompts.

Tur et al. [14] focused specifically on fine-grained retail product recognition using vision-language models. Their study is particularly relevant to this project because it tests whether general-purpose vision-language models can recognize visually similar retail products. They introduced the MIMEX dataset with 28 product categories and found that zero-shot vision-language model (VLM) performance, meaning performance without task-specific training examples, performance was not

strong enough for reliable fine-grained product classification. They then improved performance by using CLIP and DINOv2 image embeddings with prototype-based classification. This supports an important point for this project: general-purpose models are useful for quick prototyping, but they may struggle when the task requires exact SKU-level recognition.

Bai et al. [15] introduced Qwen2.5-VL, a general-purpose vision-language model that can process images, localize objects with bounding boxes or points, and output structured information such as JSON. This type of model is useful for an early MVP because it can analyze tray photos without first building a full labelled training dataset. It can also combine image understanding with language reasoning, for example by matching visible items to an inventory list. However, compared with YOLO, this approach may be slower, dependent on API access, and less stable for repeated fine-grained SKU recognition.

Ren et al. [16] presented Grounding DINO 1.5, an open-set object detection model that can locate objects based on text descriptions. The model includes both a stronger Pro version and a faster Edge version, making it relevant to real-world deployment. For this project, Grounding DINO could be useful as a middle option between YOLO and general-purpose VLMs. It may help detect objects such as "bottle", "tray", or "person holding tray" without custom training. It could also support automatic pre-labelling before training a YOLO model. However, like other open-set models, it may not be accurate enough alone for exact SKU-level inventory recognition.

Overall, these studies suggest that YOLO-based models are better suited for fast and stable detection when the target categories are fixed and labelled data is available. In contrast, general-purpose vision-language models such as Qwen2.5-VL are more flexible and easier to test, but may be slower and less reliable for distinguishing similar inventory products. Therefore, a practical approach for this project could begin with a general-purpose vision model for early testing, then move toward a custom YOLO or hybrid detection system once enough SKU images have been collected.

2.3 Camera- and Sensor-Assisted Inventory Monitoring Systems

Recent computer vision and sensor-based inventory monitoring studies show a clear movement away from manual stocktaking and single-device counting toward multi-modal, real-time verification.

Cheng et al. [17] demonstrate how ceiling-mounted cameras, computer vision-based object detection, and a digital twin/world model can be integrated to track warehouse assets such as pallets, boxes, and forklifts across multiple viewpoints. This supports the use of existing CCTV infrastructure as a continuous inventory verification layer, particularly for detecting stock presence, location changes, and scene-level anomalies. However, vision-only systems remain vulnerable to occlusion, lighting variation, camera angle limitations, and visually similar products, which makes complementary sensing important.

Sikkandhar et al. [18] address this limitation at the shelf level by developing a piezoresistive smart-shelf sensor array that captures SKU shape and weight features, then applies a TensorFlow model for product identification and misplaced-item detection. This provides a low-cost sensor-assisted approach that can verify inventory status even when camera evidence is uncertain.

Li et al. [19] extend inventory monitoring through UHF RFID localization, using drone-mounted RFID sensing and phase-based signal information with deep learning to improve warehouse stocktaking under trajectory uncertainty.

Together, these studies suggest that reliable inventory monitoring should not depend on a single data source. Instead, practical systems can combine camera observations, shelf or weight sensor readings, and RFID signal features such as RSSI and phase angle to cross-validate inventory records across multiple independent devices.

2.4 Summary and Critical Review

The existing literature shows that YOLO-based object detection models can achieve strong performance in retail product recognition, especially when they are trained on large-scale labelled datasets. However, this creates an important limitation for this project. If the system is intended to become a replicable product for different restaurants rather than a one-off solution for a single site, training a separate YOLO

model for every restaurant would require too much time and manual effort. Each restaurant has its own SKU list, packaging types, supplier products, and storage layout. The reviewed YOLO-based studies often rely on large image databases, which makes their approach less practical for small restaurant environments where data collection and annotation resources are limited. Therefore, although YOLO is technically effective, its scalability and transferability are limited in this project context.

In contrast, general-purpose vision-language models provide a more flexible direction because they can analyze images without requiring a fully trained custom dataset for each restaurant. They can also combine visual recognition with text prompts, inventory lists, and structured outputs, which makes them suitable for tray-based inventory recording. However, the key challenge is no longer simply whether the model can recognize objects, but how to improve its accuracy, consistency, and speed in this specific inventory scenario. General VLMs may confuse visually similar SKUs, miss small items, or produce unstable results across repeated images. Therefore, improving the performance of a general visual model through prompt design, inventory-list constraints, reference images, image quality control, and human confirmation becomes a central research focus of this project.

The literature also shows that relying on a single recognition method may not be sufficient for reliable inventory monitoring. A practical system should consider multi-device verification, where different data sources support or challenge each other. For example, a Raspberry Pi camera installed near a storage shelf could provide a fixed, low-cost, and visually clean way to double-check item quantities after a tray photo has been submitted. This type of camera-assisted verification could help detect mismatches between staff-submitted records and actual shelf stock. In the longer term, other sensors such as weight sensors or RFID could also be used to strengthen inventory reliability. Therefore, this project should not only focus on model selection, but also on designing a more robust verification workflow that combines general VLM recognition with low-cost camera or sensor-based cross-checking.

3 Methodology

3.1 Research Approach

Field Research The author entered the project as an outsider to the restaurant industry. Before the field visits, the author conducted online research on restaurant inventory management and existing inventory systems, but many details of real restaurant work could not be found online because they were highly practical and context-specific. Field research was therefore necessary to understand FAN's actual inventory workflow. This approach follows an "in the wild" design logic, where technology design is informed by the real setting in which the system will be used rather than only by laboratory assumptions [20]. The author collected information through interviews with restaurant staff and on-site observations of the inventory area, staff movement, item handling, and existing recording practices.

Iterative Prototyping The making process was organised through an iterative prototyping approach. Iterative user-interface design is commonly used in HCI because prototypes can be modified, tested, and revised repeatedly as designers learn more about users and tasks [21]. In total, the author carried out three field visits and developed four prototype iterations. The project did not begin with a fixed final design. Instead, each prototype was built from the author's current understanding of the restaurant workflow, then revised after new field observations, staff conversations, and prototype testing. This meant that each iteration reflected a more developed understanding of FAN's inventory management, including the original recording process, the kitchen and bar workflow differences, and the later chat-based photo recording behaviour.

3.2 Data Collection

3.2.1 Semi-structured Interviews

Semi-structured interviews were used as a practical and efficient starting point for the project because they provide guiding questions while still allowing follow-up questions when participants raise unexpected details [22]. This made them more suitable than fully structured interviews, which would have been too rigid for an unfamiliar restaurant context, and more focused than completely open-ended con-

versations. Before the first interview, the author reviewed existing restaurant inventory management tools, such as MarketMan [23], to form an initial understanding of how digital stock recording systems are usually structured. However, the conversation with the restaurant owner showed that publicly available information could not fully explain how inventory work was actually carried out in a small restaurant.

The interview questions focused on three main areas related to the three parts of the research question. First, the author asked about the current workflow for taking an item from storage and completing an outgoing-stock record, which related to reducing manual work. Second, the author asked about FAN's existing inventory management system and whether the restaurant used software similar to MarketMan, which related to improving record accuracy. Third, the author asked about the business impact of inventory problems, including the estimated weekly loss caused by food waste, which related to stock visibility and transparency. These questions provided a basic structure for the interview, but the discussion remained flexible so that the author could follow up on practical details raised by the interviewee.

Many important details, such as staff habits, time pressure, storage layout, and informal recording behaviour, belonged to practical industry knowledge that was difficult to collect from online sources alone. For this reason, the interviews were treated as a way to learn from industry practitioners rather than only to confirm a pre-existing design idea. The author prepared general topics in advance, including current stock recording methods, common problems, and possible uses of AI assistance, but the discussion remained flexible. This allowed the system requirements to be shaped by the restaurant's real workflow instead of by assumptions made before entering the field.

3.2.2 Field Observations

Field observations were used to understand the physical and practical context of FAN's inventory workflow. The observations focused on three main areas: the inventory environment, staff movement, and practical conversations with staff during the visits.

Inventory Environment. The author examined the physical inventory environment of FAN. This included the basement storage layout, shelf arrangement, lock-

ers, fridges, wine room, desk area, stairs, CCTV camera positions, power socket locations, and the types of items stored in each area. These observations helped the author understand where inventory activities took place and whether a camera, iPad, or fixed checking station could realistically be placed.

Staff Movement and Handling Behaviour. The author observed why staff entered the inventory storage area, how they carried items, and what types of items they took. When staff entered the storage area without taking any visible item, the author asked short follow-up questions to understand other uses of the space, such as printing menus, adjusting the walk-in fridge temperature, using lockers, storing personal belongings, or changing clothes.

Practical Conversations During Observation. The field visits also included practical conversations with staff during observation. The owner explained the existing outgoing-stock recording process, the restaurant manager explained staff roles and the difference between kitchen and bar workflows, and the head chef explained the kitchen team's daily stock movement and their use of WhatsApp photos for later inventory recording. The author also spoke with regular staff members to understand how they used the storage area in daily work and what made recording stock movements difficult. These observations and conversations helped identify the gap between recorded inventory data and actual staff behaviour.

3.3 Data Analysis and Requirements Development

The analysis process was organised as a step-by-step workflow after each field visit. The purpose was to turn raw field materials into clear system requirements that could guide the next prototype [24].

Organising Field Materials. First, the author organised the field materials, including observation notes, photographs, layout information, and interview notes. These materials were collected in Notion [25] and turned into a short field report with both written notes and images, so that the practical details of the restaurant environment could be reviewed after the visit.

Summarising Interview Data. For interview-based data, the author used Notion to summarise the main points from each conversation. These summaries included the interviewee's role, the problems they described, the workflow details they mentioned, and any new design constraints or suggestions. Although a full thematic analysis could have provided a more detailed qualitative interpretation, it would have required more extensive coding, theme development, and reporting [26]. This project therefore used shorter structured summaries because the findings needed to feed quickly into the next prototype between field visits. This helped separate general impressions from specific findings that could affect the system design.

Mapping Findings into Requirements. After the field report and interview summaries were prepared, the author used mind maps to organise the findings into requirement areas. Mind maps can support knowledge construction by making relationships between ideas easier to see and review [27]. Figure 1 shows one example of this process. The mind map explored how CCTV could support inventory verification in the basement storage area. It connected the observation that staff entered the storage area to retrieve items with a possible workflow where a Raspberry Pi camera sends an entry signal, the system records a timestamp and screenshot, and the system checks whether an outgoing-stock record has been created within 30 minutes. The mind map also helped identify missing information and design risks, such as how to distinguish inbound and outbound stock actions, and what happens if a person leaves between timed screenshots. These issues were then translated into prototype requirements, including clearer action definitions, human confirmation, and the need for camera evidence to support rather than replace staff-submitted inventory records.

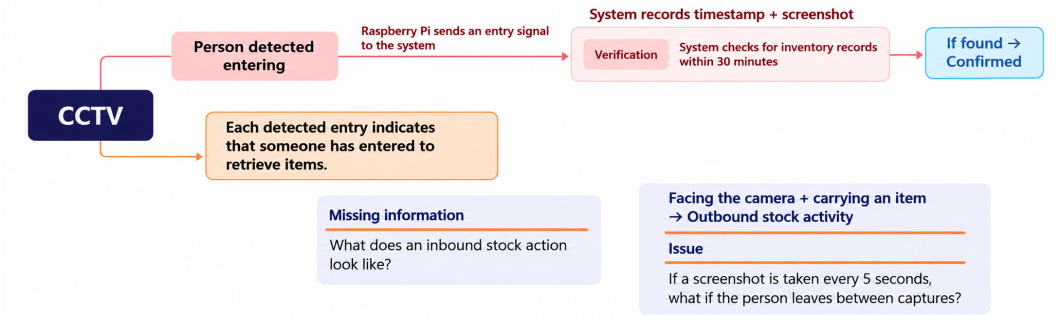


Figure 1: *Mind map used to translate CCTV-based field observations into possible system requirements.*

Using Requirements to Guide Prototyping. The final step was to use these requirements to guide prototype development. Each prototype was therefore based on the analysed field data rather than only on the author's initial assumptions. After the prototype was built and tested, the results were reviewed together with the next round of field observations, creating a repeated cycle of fieldwork, analysis, requirement definition, and prototype redesign.

4 Results

4.1 Introduction to Restaurant

4.1.1 Restaurant Background

FAN was selected as the case study site for this project [4]. It is a London-based Neo-Nikkei restaurant located in Notting Hill, combining Japanese and Peruvian cuisine with wider cultural influences. Public restaurant listings describe FAN as a higher-priced restaurant, with tasting menu options ranging from £69 to £120, and it is currently listed in the MICHELIN Guide [28]. The restaurant was founded by chef Coco Sasaki together with Romina Parra and Santiago Won, who had previously developed the FAN concept in Lima.

FAN has four counter seats and approximately 30 additional seats. This shows that FAN operates at the scale of an independent restaurant rather than a large hospitality site, making it a suitable case study for exploring inventory recording in a small but busy restaurant environment. At the same time, the restaurant handles a wide range of inventory items across kitchen ingredients, bar products, wine, and daily service stock. This combination of small scale, frequent stock movement, and operational complexity makes FAN a suitable case study for exploring whether AI-assisted inventory recording can support a busy restaurant environment without requiring a large or highly specialised technical setup.

4.1.2 Inventory Storage Layout

FAN's main inventory storage area is located in a basement level beneath the restaurant. Figure 2 shows the basic layout of this storage area. The basement includes shelves, lockers, fridges, a walk-in fridge, and a separate wine room. Staff enter the

storage area through the stairs when they need to take items for the kitchen, bar, or service.

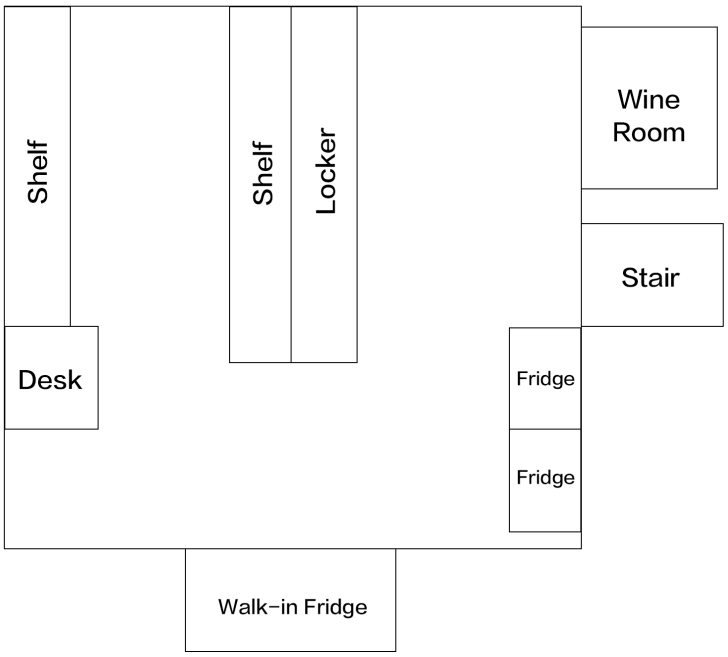
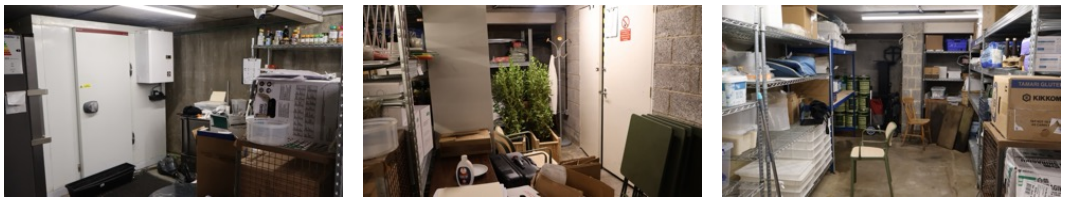


Figure 2: *Simplified layout of FAN's basement inventory storage area. Author's own diagram, not to scale.*

Figure 3 shows photographic views of the same basement inventory area. These images show the practical storage conditions observed during fieldwork, including shelving, fridge space, working surfaces, and narrow movement paths.



(a) *Fridge and shelf area* **(b)** *Storage entrance area* **(c)** *Shelf and desk area*

Figure 3: *Photographic views of FAN's basement inventory storage area.*

4.1.3 Staff Structure

Figure 4 shows the basic staff structure at FAN. The restaurant is led by the owner, Santiago, with Parasana working as the restaurant manager. Under the manager, the main operational teams are divided into the kitchen team and the bar team. The

kitchen team is led by Joel, with other kitchen staff working under this role, while the bar team consists of staff responsible for bar-related stock and service.

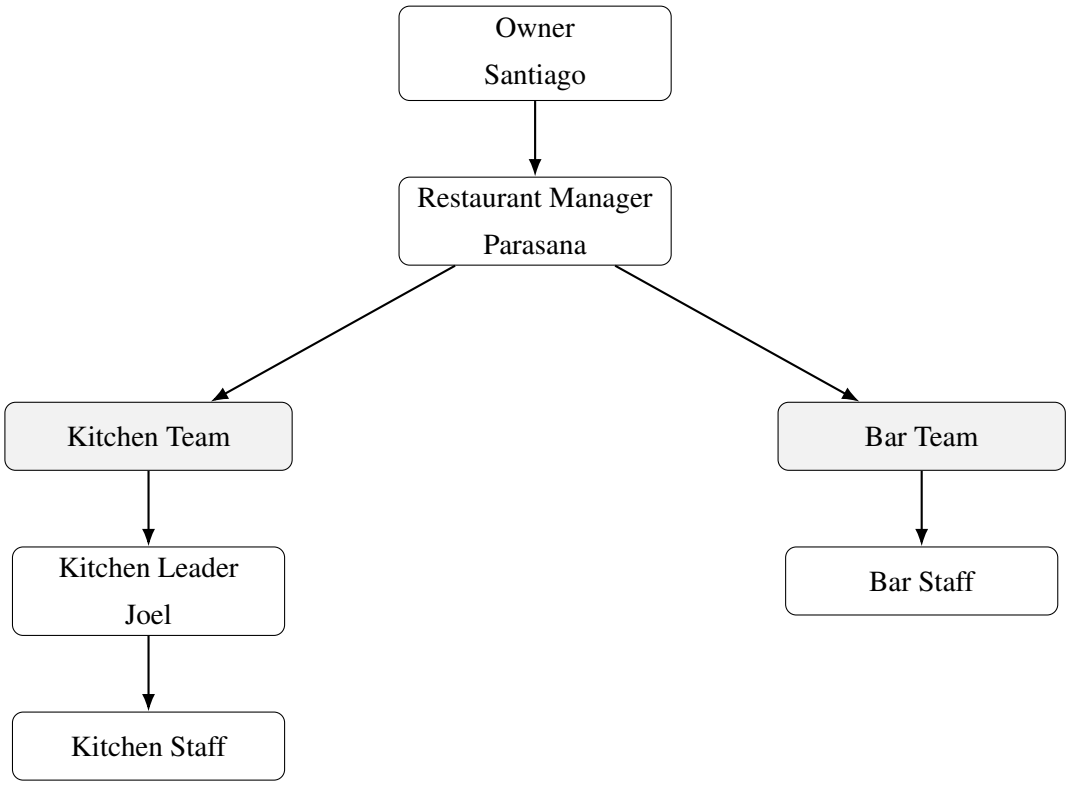


Figure 4: *Restaurant staffing structure at FAN.*

4.1.4 Current Inventory Management Process

The current outgoing-stock process at FAN is mainly manual, summarised in Figure 5. When staff need items for the kitchen, bar, or service, they go from the restaurant level to the basement storage area and collect the items they need. If the amount is small, they usually carry the items by hand. If there are several items, they use a white tray to carry them back to the restaurant.

After returning to the restaurant, staff record the movement through an internal outgoing-stock webpage. They need to search for the item name, enter the quantity taken, and upload a photo. However, the photo is not automatically checked by the system, and each record only supports one item type. This means that taking several different items requires staff to repeat the same recording process several times, which can be slow during busy service periods.

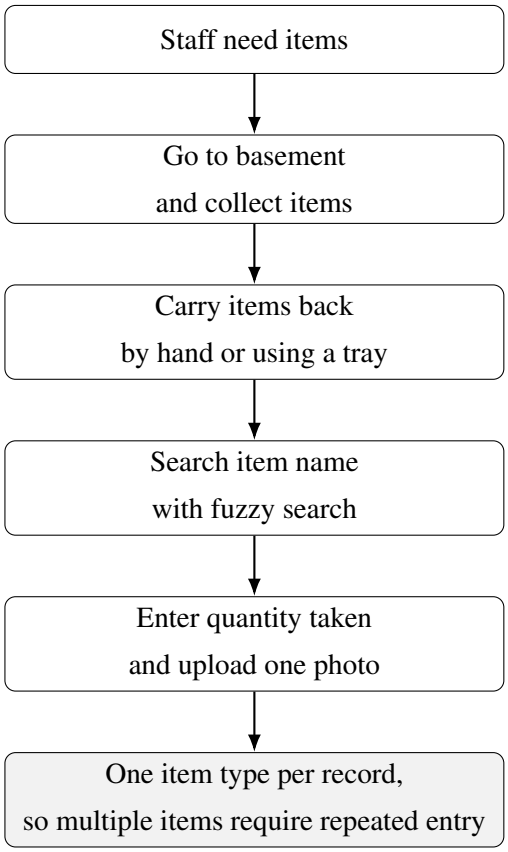


Figure 5: *Current outgoing-stock recording process at FAN.*

4.1.5 SKU Characteristics

FAN's inventory list contains 506 active stock keeping unit (SKU) records. The SKU list is not limited to food ingredients. It also includes drinks, bar stock, crockery, cleaning supplies, personal protective equipment, and other service-related items. Based on the product type field, the largest group is food ingredients, with 225 SKUs. This group includes vegetables, seafood, fish, meat, dairy products, sauces, rice, oil, spices, and Japanese, Peruvian, Chinese, Korean, and other imported ingredients. This is followed by equipment and service supplies with 99 SKUs, beverage and bar stock with 91 SKUs, cleaning and PPE with 62 SKUs, and family meal or staff food items with 27 SKUs.

This shows that the inventory problem at FAN is broader than simply recognising ingredients. The system needs to support different kinds of stock used by different teams, including kitchen items, bar items, wine, cleaning products, and service supplies. The SKU names also vary in format. Some names include product brands,

package sizes, units, or supplier-style descriptions, which makes manual search and image-based recognition more difficult. In addition, only around half of the SKU records include image URLs, meaning that the existing product list does not provide complete visual reference data for every item.

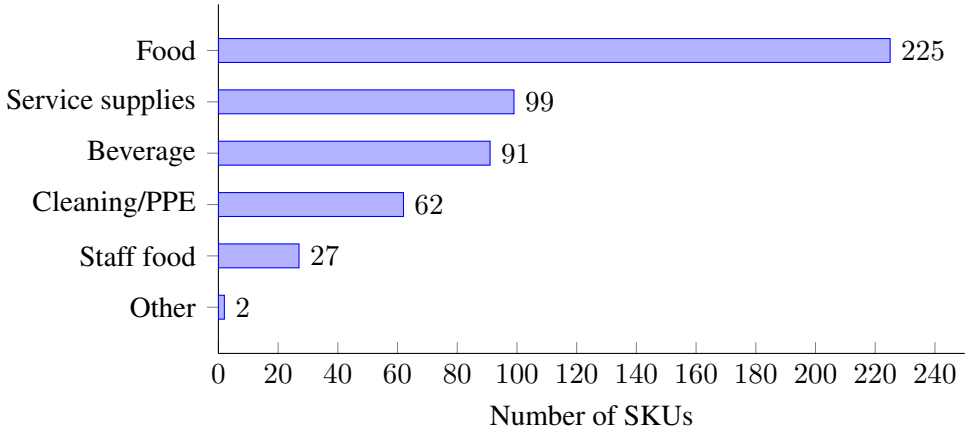


Figure 6: *Grouped distribution of inventory SKUs in FAN's product list.*

Table 1 provides an overview of the four prototype iterations and the corresponding result sections. The table is used as a guide to the development process, rather than as the final evaluation of the system.

Section	Iteration	Main aim	Observed result or design decision
Section 4.2	Iteration 1	Test full-list image recognition using a vision-language model	Response time was often more than 35 seconds, and recognition was unstable for visually similar items
Section 4.3	Iteration 2	Test whether local YOLO recognition could improve speed and reduce internet dependence	Local detection was faster, but required labelled images for each SKU
Section 4.4	Iteration 3	Reduce the recognition task using kitchen and bar contexts	Shorter candidate lists improved response time and kept human confirmation in the workflow
Section 4.5	Iteration 4	Align the system with existing staff photo-sharing behaviour	The final design became a chat-based inventory agent

Table 1: *Overview of prototype iterations and corresponding result sections.*

4.2 Iteration 1: Single-Image Inventory Recognition

4.2.1 Requirements

The first prototype aimed to test whether a restaurant employee could take one photo of inventory items and use an AI model to identify what products were shown in the image. Based on the interview with Santiago, staff commonly use a white tray to carry several items from the basement storage area back to the restaurant. Therefore, the first test scenario used a white tray containing six selected inventory products, as shown in Figure 7. The requirement was for the system to accept one image of the tray, recognise the visible item or items, and return matching inventory names from the restaurant's stock keeping unit (SKU) list.



Figure 7: *Tray-based test image used in the first image-recognition prototype.*

4.2.2 System Design

This version used a simple image-based recognition workflow. It was tested through a lightweight image-upload interface rather than a command-line-only script. The user uploaded or captured one photo, and the image was sent to a vision-language model for recognition. The restaurant's SKU list was included in the prompt so that the model could choose item names that matched the existing inventory data, rather than returning only general descriptions.

Figure 8 shows the first mobile interface concept, where recognised items were displayed for staff checking before recording.

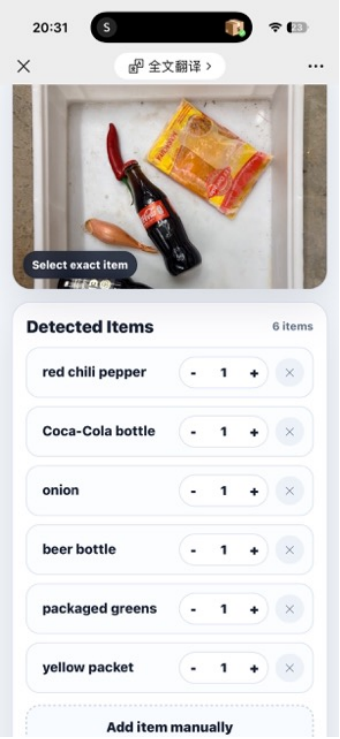


Figure 8: *Mobile interface concept for the first image-based recognition prototype.*

4.2.3 Technical Details

The prototype used Qwen3-VL-Flash as the vision-language model [29, 30]. The image and the full SKU list were included in the input prompt, so the model needed to inspect the image and compare it against more than 500 possible inventory item names. The prompt asked the model to select the most likely matching SKU from this list rather than returning a general object description. The recognised result was then displayed to the user for checking before it could be recorded.

4.2.4 Results

The result of this first prototype was unstable. In some cases, the model correctly recognised the item in the image, but in other cases it selected the wrong SKU or produced inconsistent results for visually similar products. The response time was also too long for practical use in a restaurant workflow, often taking more than 35 seconds to return a result as the model needed to compare the image against more than 500 possible SKU names each time. This delay would make the system difficult for employees to use during busy service periods, where waiting for each

recognition result is not realistic.

4.3 Iteration 2: Local YOLO-Based Version

4.3.1 Requirements

The second iteration was based on two issues identified earlier: the first prototype had a recognition response time that was too long for practical use, and the basement storage area had weak internet access. First, the system needed to improve recognition speed. The recognition time should not be longer than the time staff would normally spend entering the item manually, so the target was to complete item recognition within approximately 10 seconds. Second, the system needed to work with limited network access. Since the basement storage area had no reliable Wi-Fi or mobile signal, depending on a cloud-based model would make the system unstable in practice. Therefore, this iteration explored whether a local model could support faster recognition without relying on a constant internet connection.

4.3.2 System Design

The design idea was to move the recognition workflow closer to where items were stored. Since there was a desk in the basement storage area, the proposed setup was to place an iPad or similar device on the desk. Staff could take items from the storage shelves, bring them to the desk, and use the device to recognise the items before taking them back upstairs.

In this workflow, the recording action would happen inside the basement rather than after staff returned to the restaurant level. This was intended to reduce the chance that staff would forget to record items after leaving the storage area.

4.3.3 Technical Details

This iteration used YOLOv8n as the starting point. YOLOv8 [31] is a real-time object detection model designed to identify and locate objects in images using bounding boxes. The nano version, YOLOv8n, is the smallest model for YOLOv8. It was selected because it is lightweight and designed for faster inference with lower computational cost, which matched the aim of running recognition locally in the basement storage environment.

The pretrained YOLOv8n model was trained for general object detection and could recognise common object categories such as people, bottles, cups, bowls, chairs, and some food items. However, these categories are generic and do not match FAN's specific stock keeping unit (SKU) names, such as exact drink brands, sauce types, wine labels, or packaged restaurant products. Therefore, custom fine-tuning was required for FAN-specific inventory recognition.

As a proof-of-concept test, the first training experiment focused on a single SKU rather than the full product list. Coca-Cola was selected because it is a common packaged product with a clear visual appearance. A dataset of 50 Coca-Cola images was collected and manually annotated with bounding boxes. This number was chosen as a practical upper limit for a lightweight training workflow, as collecting and annotating more images for a single item would reduce the efficiency benefit of the system.

The annotated images were then used to develop a YOLOv8n proof-of-concept model with Coca-Cola as a custom object class. The aim was to test whether a locally trained object detection model could recognise a restaurant inventory item quickly under the controlled tray-based setup described above. Figure 9 shows an example output from this proof-of-concept test.

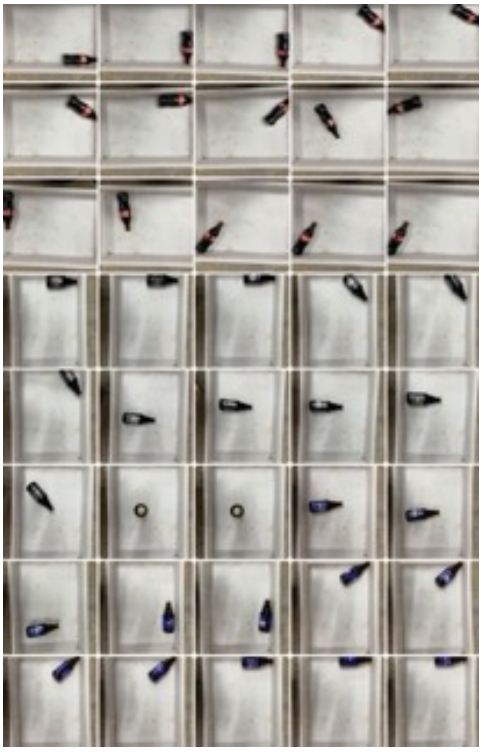


Figure 9: *YOLOv8n Coca-Cola proof-of-concept detection result.*

4.3.4 Results

This iteration addressed the speed and network-dependence problems identified in the first prototype. A local YOLO-based approach had the potential to reduce recognition time and avoid the need for Wi-Fi in the basement. However, it also introduced a new challenge. Training a YOLO model required labelled image data for each inventory item, and FAN's SKU list contained many different products with varied packaging, sizes, and visual appearances.

This made the approach difficult to scale. For the system to recognise many restaurant items reliably, a large number of labelled training images would be needed. This was not practical within the time and data collection limits of the project. As a result, this iteration showed that although local object detection could improve speed, it was not the most suitable approach for a flexible restaurant inventory system with many changing SKUs.

4.4 Iteration 3: Fieldwork-Informed Recognition Workflow

4.4.1 Requirements

The requirements for the third iteration changed after the second field visit to FAN. Two observations were especially important. First, kitchen staff and bar staff used different parts of the inventory. Kitchen staff mainly took kitchen ingredients and food-related items, while bar staff mainly took drinks, wine, and bar products. This meant that each recognition task did not need to match the image against the full SKU list. Instead, the system should separate the recognition workflow into kitchen and bar contexts, so that each mode only uses the most relevant part of the inventory list.

Second, the tray-based basement workflow was less practical than expected. In discussion with the restaurant manager, who had a better understanding of daily restaurant operations, it became clear that staff usually try to avoid using trays when possible. The kitchen space is small, and there is limited space to place a tray. Carrying a tray also creates extra work because staff may need to make another trip to return it. Therefore, the recognition task should happen in the kitchen rather than at a fixed checking station in the basement. This also made the system more practical because the kitchen had better Wi-Fi and mobile signal than the basement storage area.

To test this requirement, two controlled test images were added, as shown in Figure 10. One image contains only beverage-related products and was used for the bar mode test. The other image contains only food-related products and was used for the kitchen mode test.



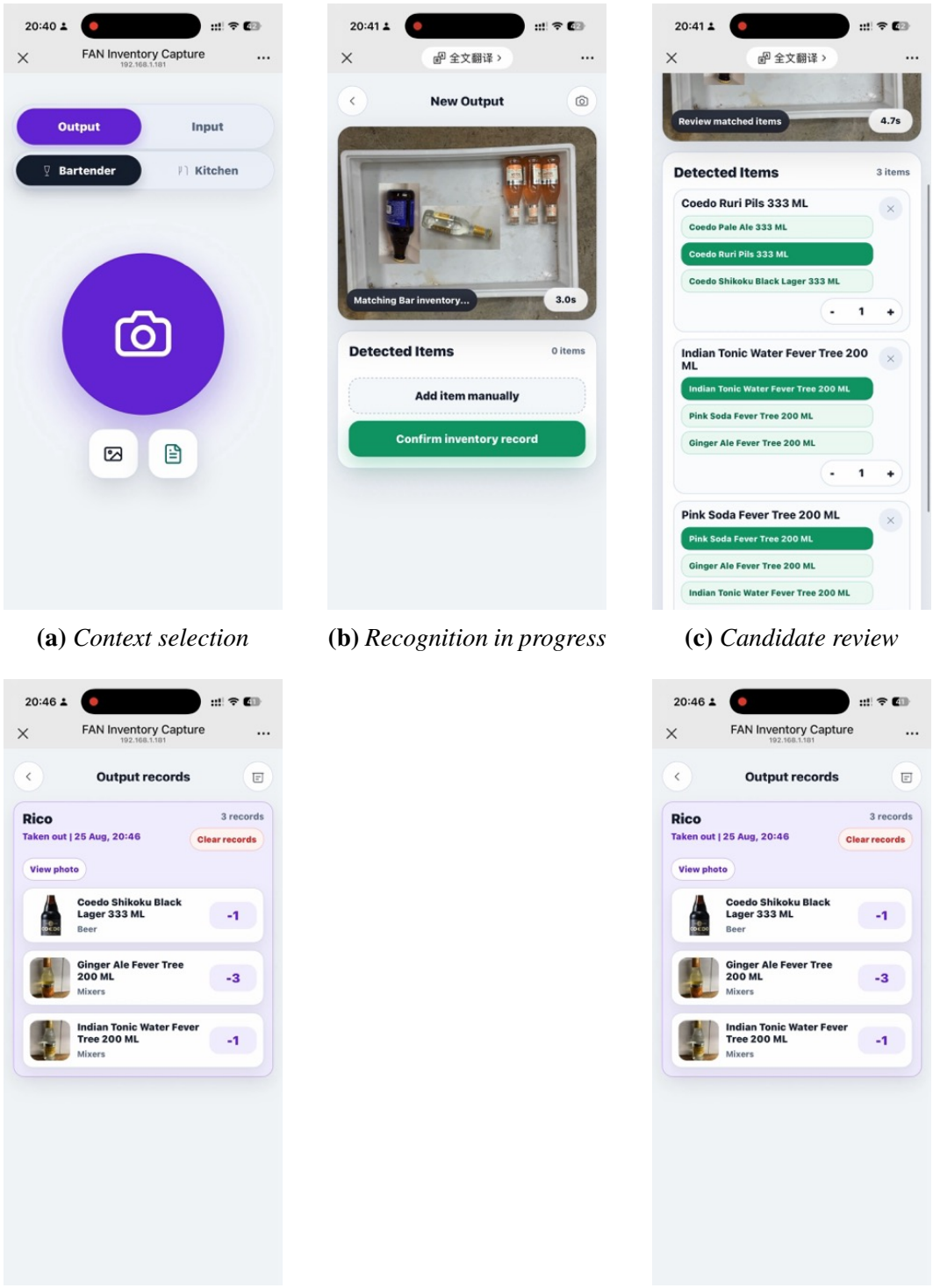
Figure 10: *Controlled test images for bar-mode and kitchen-mode recognition.*

4.4.2 System Design

Based on this observation, the system was redesigned with two modes: kitchen mode and bar mode. In kitchen mode, the model only needed to compare the image against a shorter list of kitchen-related items. In bar mode, the model only needed to compare the image against a shorter list of bar and drink items. The lists were further reduced using recent outgoing-stock records, so commonly used items from the previous week were prioritised.

The interface was organised as a small web-based testing screen with a kitchen/bar mode selector, an image upload area, and a result panel for the recognised item. This version also introduced a human-in-the-loop correction step. The model returned a suggested item name, but the result was not saved automatically. Staff could check the suggestion before confirming it. If the recognised item was incorrect, the interface provided several similar or easily confused items as alternatives, allowing staff to quickly select the correct one. This design reduced the risk of saving incorrect records while keeping the correction process fast.

Figure 11 shows the main interface states for this iteration, from selecting the inventory context and submitting a photo to reviewing matched items, viewing record evidence, and saved output records.



(a) *Context selection* **(b)** *Recognition in progress* **(c)** *Candidate review*

(d) *Record evidence* **(e)** *Saved records*

Figure 11: *Interface flow for the fieldwork-informed recognition workflow in Iteration 3.*

4.4.3 Technical Details

This version returned to the Qwen vision-language model, but the recognition task was made smaller and more focused. Instead of sending the full inventory list to the model, the system selected a shorter SKU list according to the chosen mode. If kitchen mode was selected, the prompt contained kitchen-related items. If bar mode was selected, the prompt contained bar-related items.

The SKU list was also filtered using recent stock movement history. Items that appeared more frequently in recent outgoing-stock records were prioritised in the recognition list. This reduced the number of candidate items and made the prompt shorter. The aim was to improve the recognition workflow without training a custom object detection model.

4.4.4 Results

This iteration produced a clear improvement in both response time and recognition accuracy. Because the model only needed to compare the image against a smaller and more relevant SKU list, the response time was reduced to around 5 seconds in most cases, and generally stayed below 10 seconds. This made the recognition process faster than the earlier full-list Qwen prototype and closer to the time needed for manual entry.

The recognition accuracy also improved. In most cases, the system was able to identify the correct item from the reduced kitchen or bar list. Errors mainly occurred when products were visually very similar, such as different flavours of Fever-Tree drinks or items with similar packaging. However, these mistakes could usually be corrected manually within around 3 seconds, because the interface allowed the user to edit the recognised item before saving the record.

Overall, this iteration showed that narrowing the recognition context made the vision-language model approach more practical. It did not remove the need for human confirmation, but it reduced both the waiting time and the effort required to correct recognition errors.

4.5 Iteration 4: Chat-Based Inventory Agent

4.5.1 Requirements

The requirements for the fourth iteration changed after the third field visit to FAN. Although the previous version was already a workable system, the fieldwork showed that the system could be better aligned with the kitchen team's actual working habits. During a conversation with the head chef, the author found that the kitchen team collected items from the basement storage every day, rather than only three times per week as suggested by the outgoing-stock records. The head chef also explained that the number of commonly used kitchen items was larger than expected. This meant that the system still needed to support a wider range of kitchen ingredients and daily stock movements.

Another important observation was the use of an existing WhatsApp group, as shown in Figure 12. Staff already used this group to send inventory photos, so that missing records could be completed later when they had time. These photos were usually clear single-item images of ingredients, as shown in Figure 13. This changed the design requirement of the system. Instead of asking staff to place several items on a tray or use a separate checking station, the system could follow an existing behaviour and work as a chat-based recognition agent. The new requirement was therefore to test whether recognising one clear item image at a time could improve response speed and accuracy.

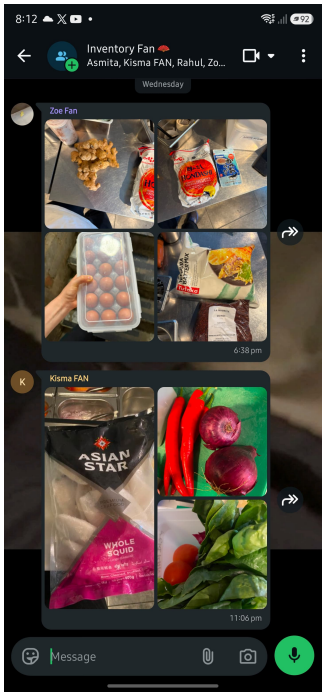


Figure 12: *Existing WhatsApp group used by staff to share inventory photos.*



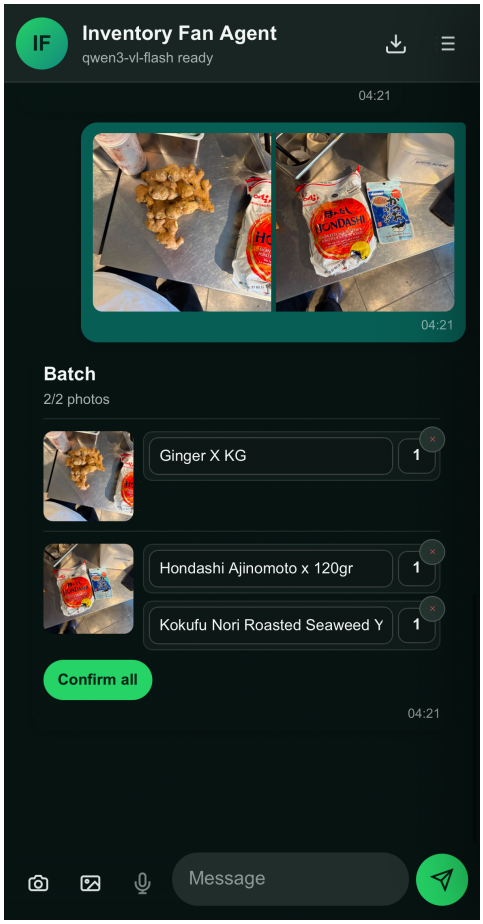
Figure 13: *Examples of clear single-item inventory photos shared by staff.*

4.5.2 System Design

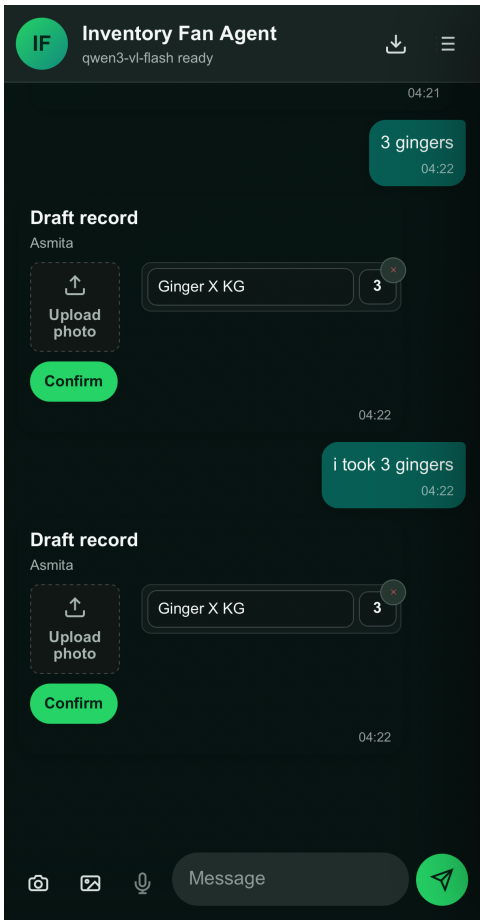
Iteration 4 redesigned the system from a separate image-upload tool into a chat-based AI inventory agent. The interface was designed to look similar to WhatsApp, because staff at FAN were already familiar with this type of communication. This reduced the learning cost and made the system closer to their existing working habits.

In this version, staff can upload one or multiple inventory photos directly in the chat interface. After receiving the images, the agent analyses them and returns a proposed list of outgoing-stock record entries. Staff can then review the list and confirm it with one action. The records are only saved after staff confirmation, so the system remains human-in-the-loop and does not automatically add uncertain results to the inventory database, as shown in Figure 14.

The agent also supports natural-language input through typed or voice messages, shown in Figure 14. For example, staff can type or say “2 ginger” or “I took 2 gingers”, and the agent converts this message into a structured outgoing-stock entry. If a photo is not available at that moment, staff can still create the record first and upload the supporting image later. This design allows the system to support both image-based recognition and quick manual reporting within the same chat workflow.



(a) *Image-based recognition from uploaded inventory photos*



(b) *Text-based recognition from natural-language input*

Figure 14: *Recognition workflows in the chat-based inventory agent*

4.5.3 Technical Details

The fourth iteration still used the Qwen3-VL-Flash vision-language model [29, 30]. However, the recognition workflow was changed from direct SKU selection to a two-step matching process. Instead of sending the full SKU list to the model, Qwen was first asked to return a short visible item name, such as “ginger” or “hondashi”. The application then matched this returned name against the local SKU list. This reduced the prompt length and moved the exact SKU matching step into the local interface, which helped shorten the response time.

The local SKU matching step was implemented as a lightweight rule-based scoring system in JavaScript, rather than as another AI model. Each SKU was converted into a searchable text string using its item name, type, tags, and keywords. If the

CSV file already contained keywords, these were used directly. Otherwise, the system generated simple keywords from the SKU type and item name. For example, soda or mixer items were given keywords such as “soft drink”, “bottle”, and “can”, while vegetable or fruit items were given keywords such as “fresh produce”. Specific item names also triggered additional keywords, such as “ginger root” for ginger and “dashi stock” for hondashi.

The item name returned by Qwen was also normalised and split into search tokens. The matcher then ranked possible SKU candidates based on token overlap, whether the token appeared directly in the SKU name, historical frequency, and category-based weighting. Additional rules were added for fresh ingredients. For example, the word “ginger” could refer to either the kitchen ingredient “Ginger X KG” or bar products such as ginger ale and ginger beer. To reduce this error, the matcher gave higher scores to kitchen and fresh-produce items when the returned item looked like a fresh ingredient, and reduced the score of bar or drink-related SKUs. The final matched item was still shown as a draft record, so staff could review and correct it before saving.

4.6 Evaluation of Final Design

The final evaluation shows that the system is fast and performs well in image recognition across different types of inventory items. However, the recognition accuracy is also influenced by how the images are taken.

Table 2: *Final Evaluation Results*

Test	Input	Items	Expected SKU(s)	Result	Time
1	Single photo	1	Ginger	Correct	0.2 s
2	Single photo	2	Hondashi Ajinomoto / Yakitori	Correct	0.2 s
3	Single photo	3	Pink Soda Fever Tree 200ML	Correct	1.3 s
4	Single photo	1	Red Chili Pepper	Incorrect: matched Momoya Seasoned Oil Layu with Chili	0.3 s
5	Single photo	1	Squid Whole 11/20 × 800gr	Correct	0.6 s
6	Single photo	4	Red Chili Pepper; Spinach; Ginger Beer Fever Tree 200ML; Maracuya Packet	Partial: incorrect matches for Red Chili Pepper and Maracuya Packet	1.2 s
7	Single photo	1	Maracuya Con Pepas × 250gr	Correct	0.3 s
8	Single photo	1	Coedo	Incorrect: matched Ginger Beer Fever Tree 200ML	0.3 s
9	Single photo	1	Coca Cola 300ML	Correct	0.3 s
10	Single photo	5	Coca Cola 300ML; Hazelnut Syrup; Kwai Feh Lychee Liqueur; Onion Red; Red Chili Pepper	4/5 matched; Hazelnut Syrup not matched because it was not present in the SKU database	1.1s
11	Four photos	6	Multiple SKUs	Correct	3.9 s

In terms of speed, recognition of a single image was generally completed within **one second**. When multiple images were submitted together, the processing time increased approximately in proportion to the number of images. This indicates that the system can provide fast feedback during practical inventory recording.

In terms of accuracy, the system performed best when the product label was

facing the camera, the image was clear, and important text on the packaging was visible. Images containing only one or two different items also generally produced more reliable results than images containing many different products.

The arrangement of items can also affect quantity recognition. When multiple units of the same product are included in one image, such as four chilli peppers, placing them separately makes it easier for the system to identify the correct quantity. Overlapping or closely grouped items may lead to inaccurate counting.

The evaluation also revealed that some products were not included in the provided SKU list. For example, the Hazelnut Syrup in the evaluation could be identified from the image but could not be matched to an SKU because no corresponding item existed in the database. This suggests that further collaboration with the restaurant is needed to complete and maintain the SKU list, ensuring that products used in daily operations are properly represented in the system.

Overall, the evaluation suggests that the system performs well across the tested image recognition scenarios. At the same time, recognition performance is influenced by how images are taken and whether the relevant products are included in the SKU database. Clearer image capture practices and a more complete SKU list could therefore further improve the reliability of the system.

5 Discussion and Conclusion

This section interprets the prototype results in relation to the research question and the literature review. It discusses how the final design addresses manual work, record accuracy, and stock transparency, while also recognising the limits of this small case-study project.

5.1 Answer to Research Question

Reducing manual work. The project suggests that manual work can be reduced by designing the recording process around the way staff already handle inventory, rather than asking them to follow a separate technical workflow. The original outgoing-stock system required staff to search for each item, enter quantities one at a time, and upload a supporting photo for each record. This created repeated work during busy service periods. The final chat-based prototype reduced this friction

by allowing staff to submit photos or short messages in one familiar interface and review the resulting draft records together. This means the system does not replace staff judgement, but it reduces repeated data-entry steps.

Improving record accuracy. The results show that record accuracy depends on both model performance and workflow design. The first prototype showed that asking a vision-language model to choose from the full SKU list was slow and unstable, especially when products looked similar. The local YOLO proof of concept showed that custom object detection could improve speed, but only if enough labelled images were collected for each item. The later prototypes therefore used a more practical compromise: reducing the candidate list by kitchen or bar context, using recent stock movement history, and asking staff to confirm or correct draft records before saving. This human-in-the-loop workflow is important because AI recognition can support accuracy, but should not be treated as a fully automatic source of truth in a fine-grained restaurant inventory task.

Supporting more transparent stock management (RQ3). The system can support transparency by connecting each outgoing-stock record with visible evidence and an understandable decision process. In the final prototype, the staff-submitted image, the model's visible item description, the matched SKU, and the confirmed quantity are kept together in the recording flow. This makes the record easier to review than a manual entry that only stores a typed item name and quantity. The literature review also suggests that future inventory systems could become more transparent by combining multiple verification sources, such as submitted photos, fixed shelf cameras, weight sensors, and RFID data.

5.2 Limitations

The main limitation of this project is that it was based on a single restaurant case study. FAN has its own storage layout, staff structure, menu style, and inventory list, so the findings cannot automatically be generalised to all restaurants. The project shows that the proposed workflow can fit this specific environment, but a wider study would be needed to understand how well it transfers to other restaurant types, such as large chains, casual dining restaurants, cafes, or bars.

The second limitation is that the project did not include a formal user study with

questionnaires or a long-term deployment. This was partly because the project focused on understanding the workflow and building a practical MVP within a limited timeframe. As a result, the evaluation is based on prototype testing, field observations, and design reflection rather than statistically robust user-satisfaction data.

The third limitation is the lack of an independent double-check mechanism for inventory records. Currently, the system mainly relies on staff to record stock movements. In future work, CCTV could be used as an additional verification layer by detecting whether staff enter or leave the storage area and comparing these activities with the inventory records submitted by staff. This would not replace staff input, but could provide an additional layer of verification and make the inventory process more transparent.

Finally, the evaluation revealed several operational and management issues that need to be addressed before wider deployment. For example, some products used by the restaurant were not included in the provided SKU list, meaning that the SKU database still needs to be completed and regularly updated in collaboration with the restaurant. Staff training is also necessary, particularly on how to take images that are suitable for recognition, such as keeping the image clear, making product labels visible, and separating multiple items when possible. These issues show that successful deployment depends not only on the recognition technology itself, but also on maintaining accurate inventory data and establishing clear working practices for staff.

5.3 Future Work

Future work should first evaluate the system in a wider range of restaurant environments. As the current project was developed and tested with FAN, testing the workflow in restaurants with different storage layouts, menus, staff structures, and inventory sizes would help understand how well the system can be adapted to other operational contexts.

A longer-term deployment with restaurant staff would also be valuable. Rather than evaluating the system mainly through prototype testing, future studies could observe how staff use the system during daily inventory tasks and collect feedback on usability, efficiency, and recognition accuracy over time. This could also help identify practical issues that may not appear during short-term testing.

Another direction is to introduce CCTV as an additional double-check mechanism. CCTV could detect staff entering or leaving storage areas and compare these activities with submitted inventory records. This would provide an additional layer of verification and improve the transparency of inventory movements without replacing staff input.

Finally, future deployment would require closer collaboration with the restaurant to maintain a complete and up-to-date SKU list and establish clear working practices for staff. This includes adding missing products to the SKU database and providing simple guidance on how to take images that are more suitable for recognition, such as keeping labels visible, taking clear images, and separating multiple items when possible.

5.4 Conclusion

This thesis explored whether AI-assisted inventory monitoring could reduce manual work, improve record accuracy, and support more transparent stock management in a restaurant environment. Through field research at FAN and several prototype iterations, the project found that a fully automatic recognition system is not the most practical starting point for small restaurant inventory work. A local YOLO model can be fast, but requires too much labelled data to scale across many changing SKUs. A general vision-language model is more flexible, but needs context filtering and human confirmation to become reliable. The final chat-based inventory agent therefore offers a practical middle path: it follows staff's existing photo-sharing behaviour, reduces repeated manual entry, and keeps the user in control of final record confirmation. While further testing is needed, the project shows that AI can support restaurant stock recording when the system is designed around real working practices rather than technology alone.

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